

- **Cooperative scheduler in C and Python** (Advanced System Software) — Implemented trap entry via mtvec, three processes with per-process saved PCs, switched by rewriting mepc inside the ecall handler.
- **Dynamic Tick Scheduler, xv6-riscv** (Advanced System Software) — Implemented a dynamic timer tick in the MIT xv6-riscv kernel, instrumenting context switches, timer activity, and process states in kernel and user space; average usertests execution ticks fell 29.5% from the static-tick baseline.
- **Cross-Demographic IMU Modelling** (Machine Learning) — Built an RNN next-step predictor for 3-axis accelerometer data (MotionSense); measured per-subject MSE degradation across age/gender groups and explained it via PCA axis loadings.
- **Multimodal Sensor Fusion on Isomap Embedding** (Unsupervised Machine Learning) — Embedded 32-D latents from a multimodal autoencoder (front-camera, LiDAR BEV, pose-derived dynamics; PandaSet) onto a 2D Isomap manifold, clustered it with K-means, and showed the clusters correspond to driving regimes via speed and LiDAR-range overlays.
- **Path Planning and PID Control, Waffle-Pi Robot** (Autonomous Systems) — Built a ROS navigation stack that plans a route with RRT and tracks it with a PID pose controller in Gazebo.
- **Neural Network Controller, Waffle-Pi Robot** (Autonomous Systems) — Trained a neural network to imitate an MPC expert driving a differential-drive robot, replacing the hand-tuned controller.
- **Gray-Coded M-QAM Link Simulation** (Digital Communications) — Implemented 4-QAM and 64-QAM modulation, AWGN transmission, and coherent demodulation in MATLAB, extending a fixed 16-QAM reference to arbitrary square constellations; benchmarked simulated bit error rates.